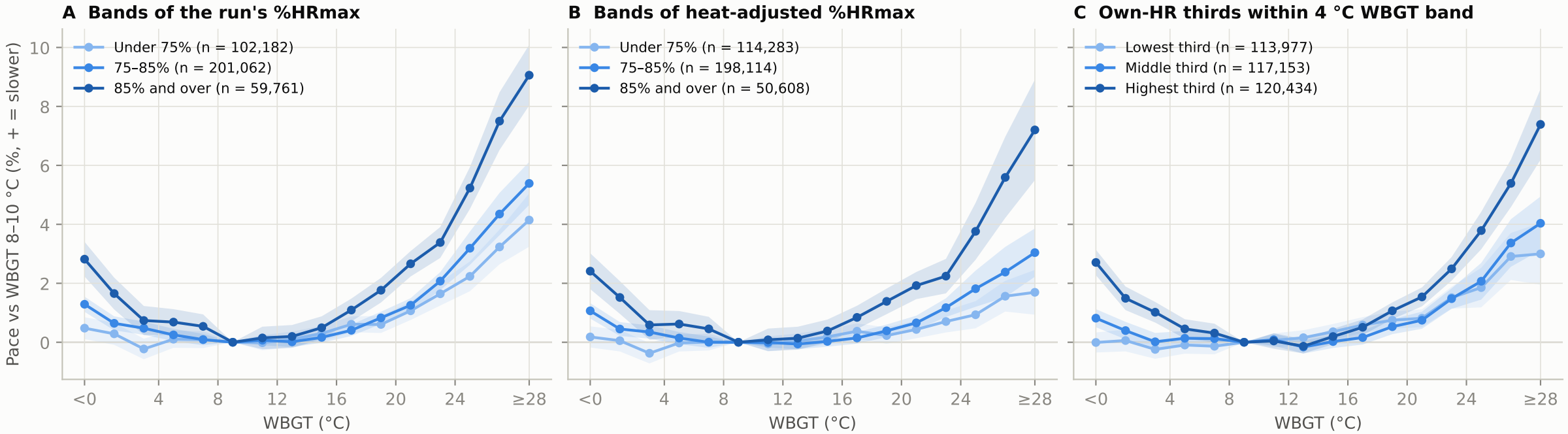

Within runner, separate model per band. A: bands from the run's mean HR. B: HR minus the average within-runner effect of the WBGT bin on HR at matched pace, then the same cut points. C: thirds of HR among the runner's runs in the same 4 °C WBGT band (runner-band cells of 6 or more runs). Controls: log distance, climb per km, 30-day fitted pace. 95% CI clustered by runner.